

Date: 4th May, 26

Sun Mon Tue Wed Thu Fri Sat

Sheikh Muhammad Ibrahim

Theory of Automata Assignment 3

F2623-630

Section A

Q₁ Define a Non-Deterministic Finite Automata (NFA)

Ans) A NFA is a finite state machine in which

① For a given input multiple next states may exist

② It may also include 'n' or 'ε' transitions.

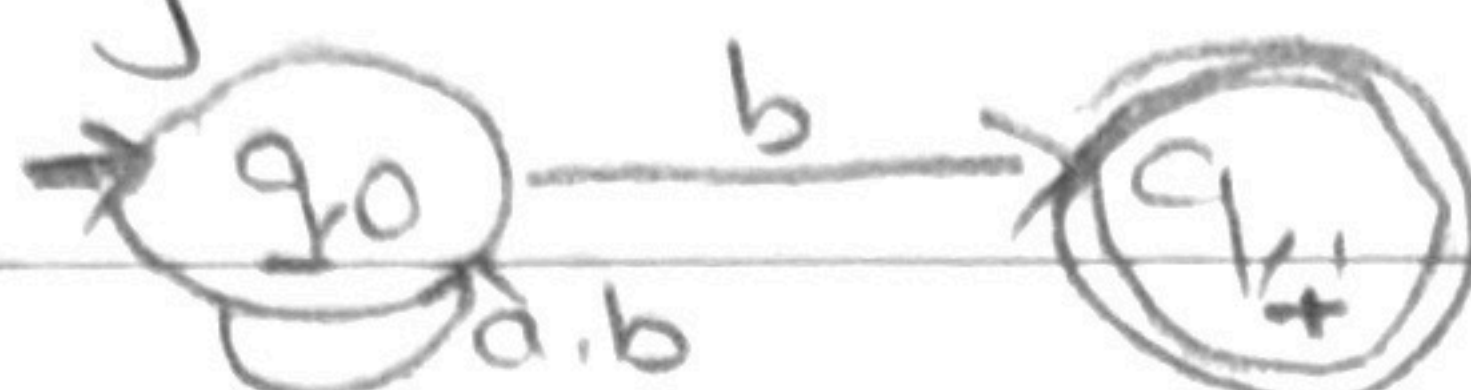
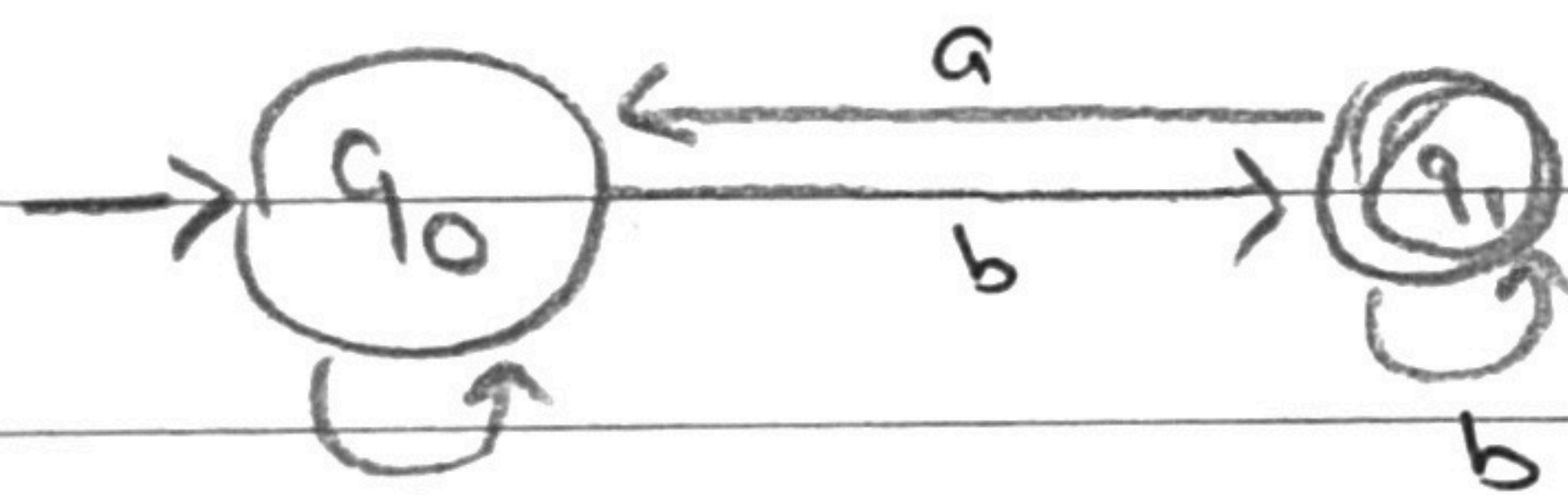


Figure 1.1

Q₂ Difference between NFA and DFA

Ans)	Feature	DFA	NFA
	Determinism	one transaction per input	Multiple or Zero transitions
	No. of Paths	One Computation path	multiple possible paths
	Acceptance	Single path must accept	any path accept
	Complexity	Easier to implement	Easier to design
	Conversion	Cannot convert to NFA	Can Convert to DFA

Figure 1.1 to DFA



Q₃ Relationship between NFA and TG

Ans) NFA is a formal mathematically model, while transition Graph is a graphical representation of automata. Every NFA can be represented as a TG but not every TG strictly follows NFA rules.

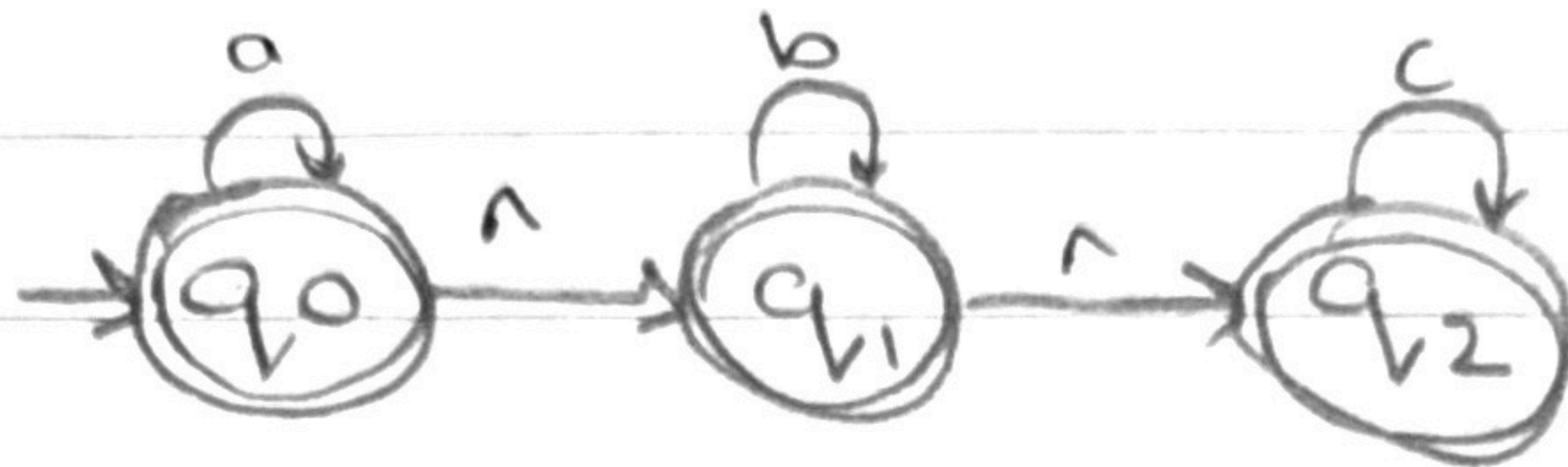
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Section B

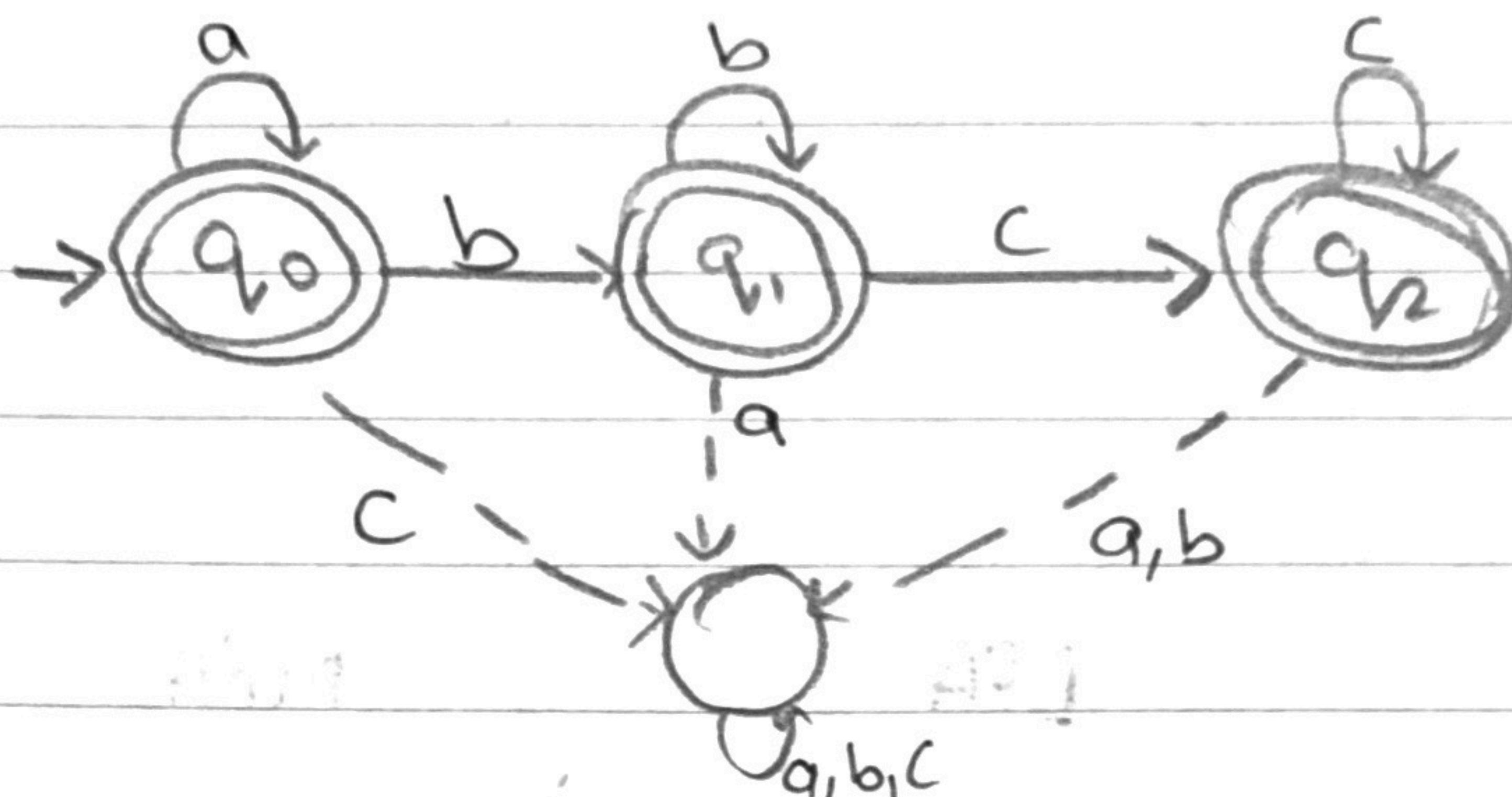
Q4) Draw both NFA and DFA.

$L_1 = \{a^*b^*c^*\}$ Alphabet: $\Sigma = \{a, b, c\}$

NFA

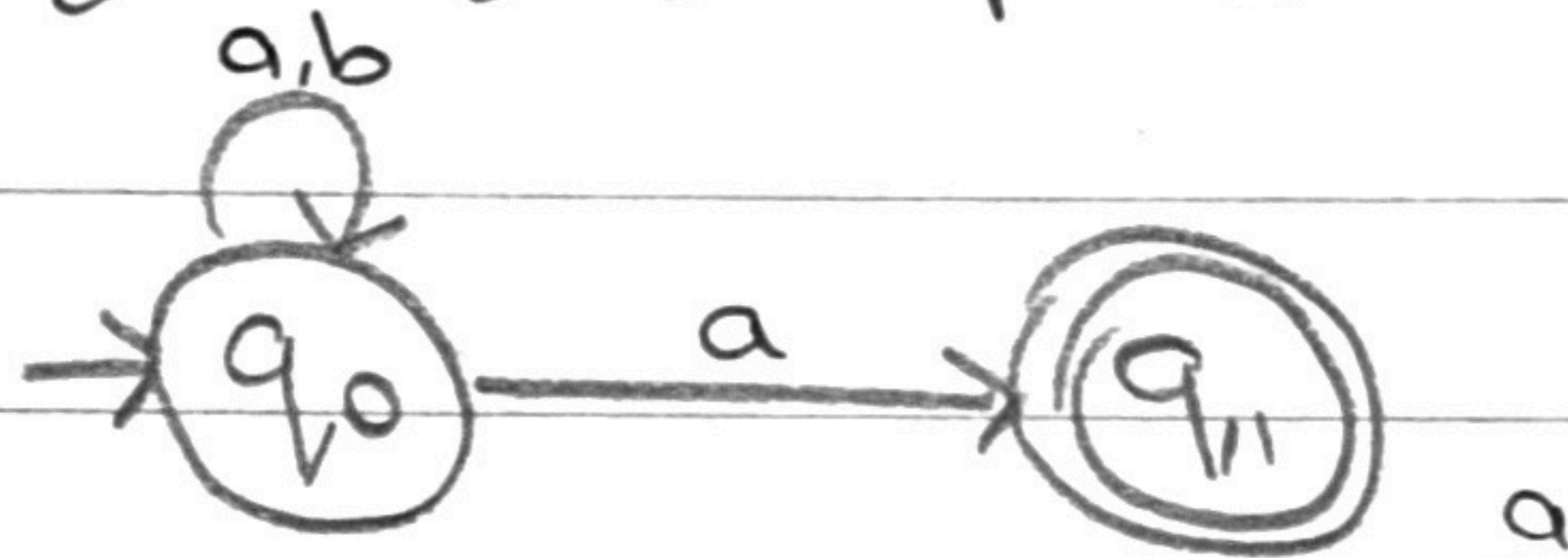


DFA

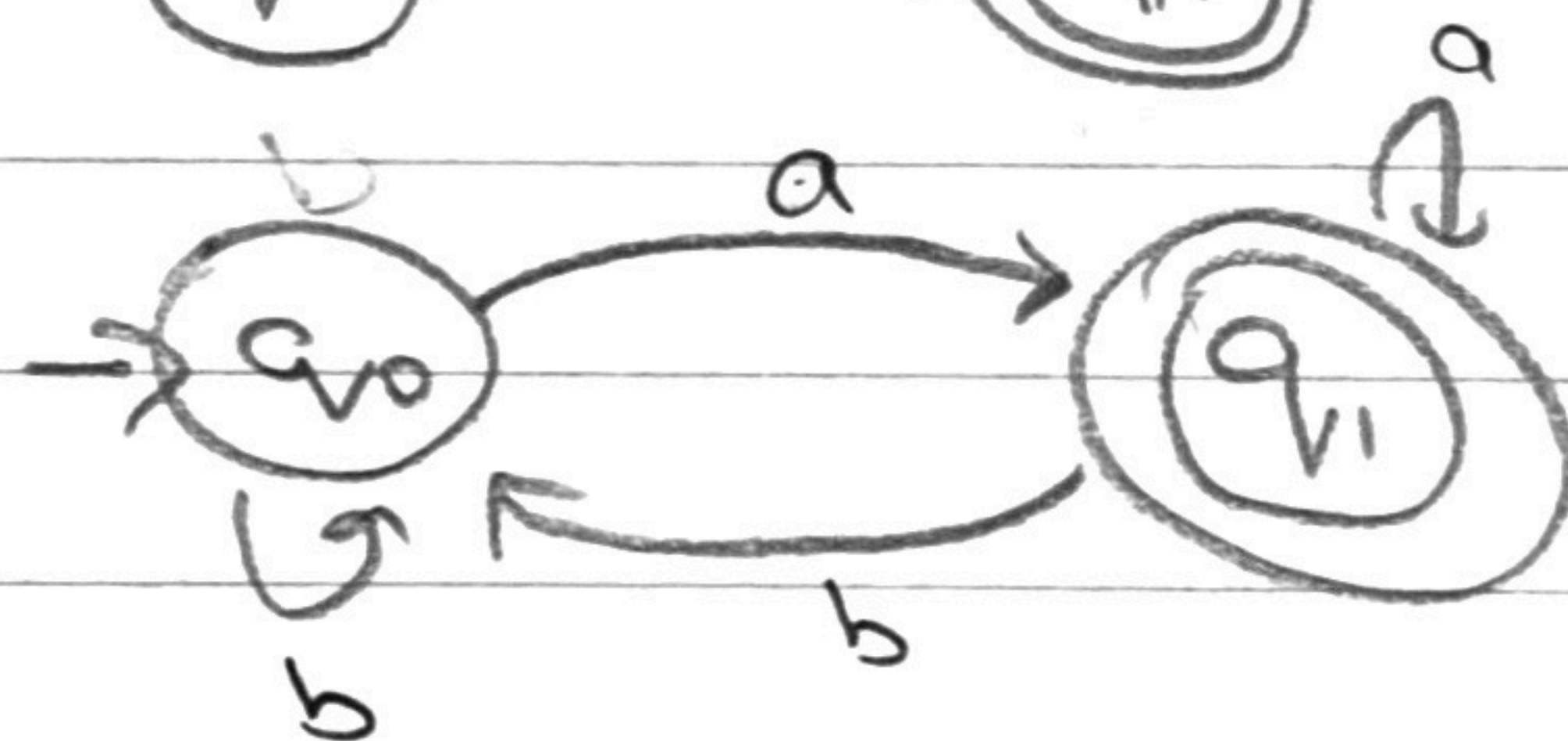


Q5) $L_2 = \{(a+b)^*a\}$ Alphabet: $\Sigma = \{a, b\}$

NFA

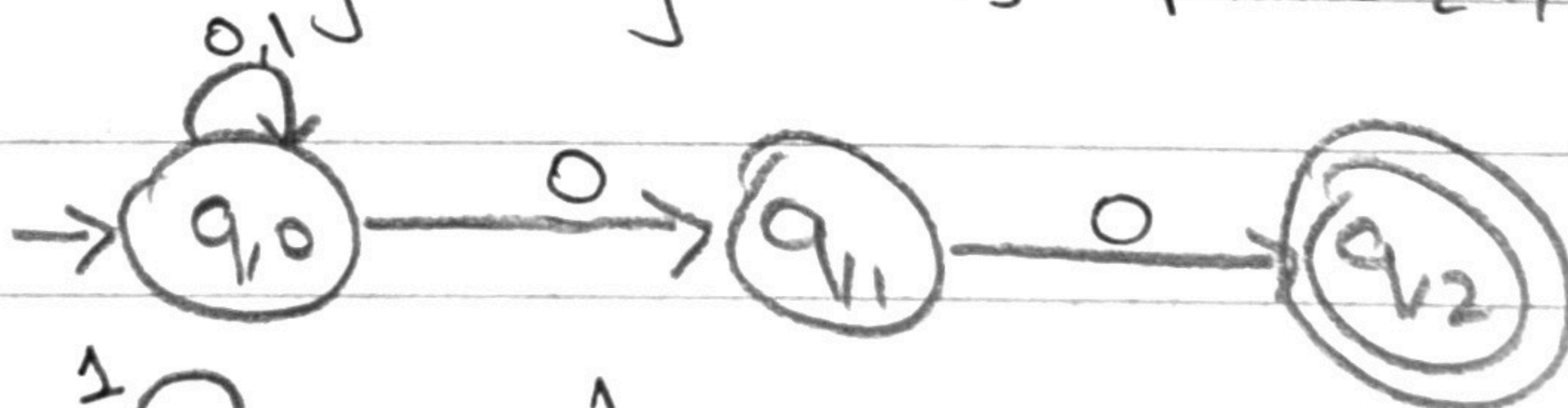


DFA

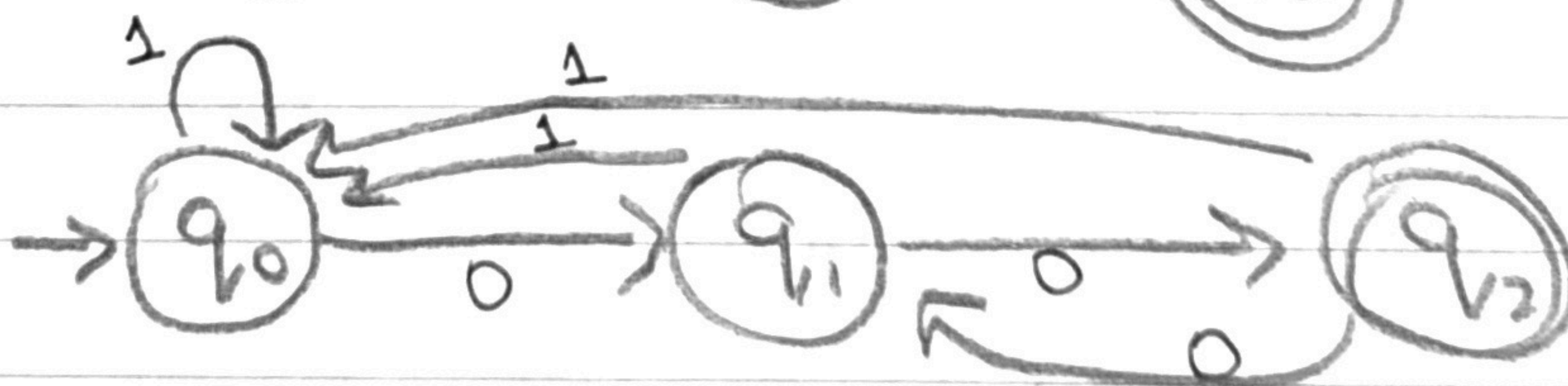


Q6) $L_3 = \{\text{All strings ending with } 00\}$ Alphabet: $\{0, 1\}$.

NFA



DFA

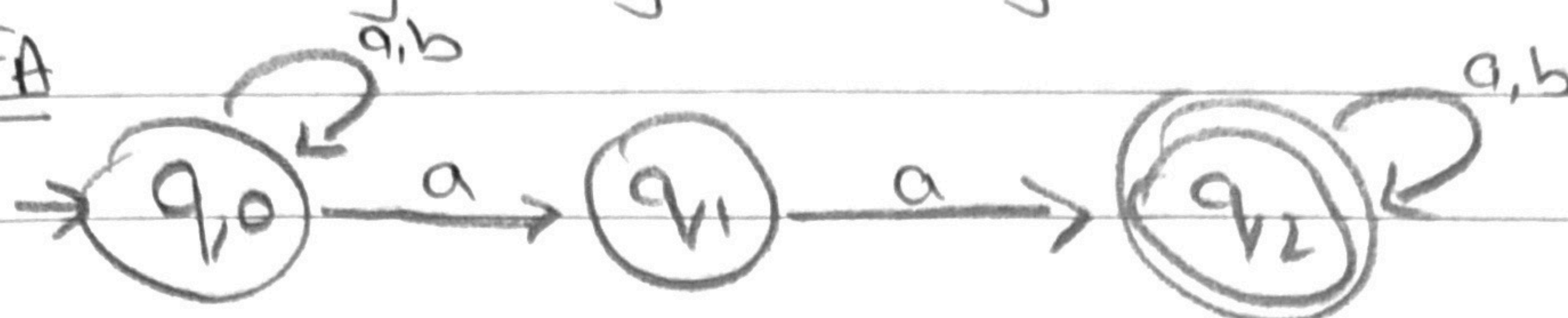


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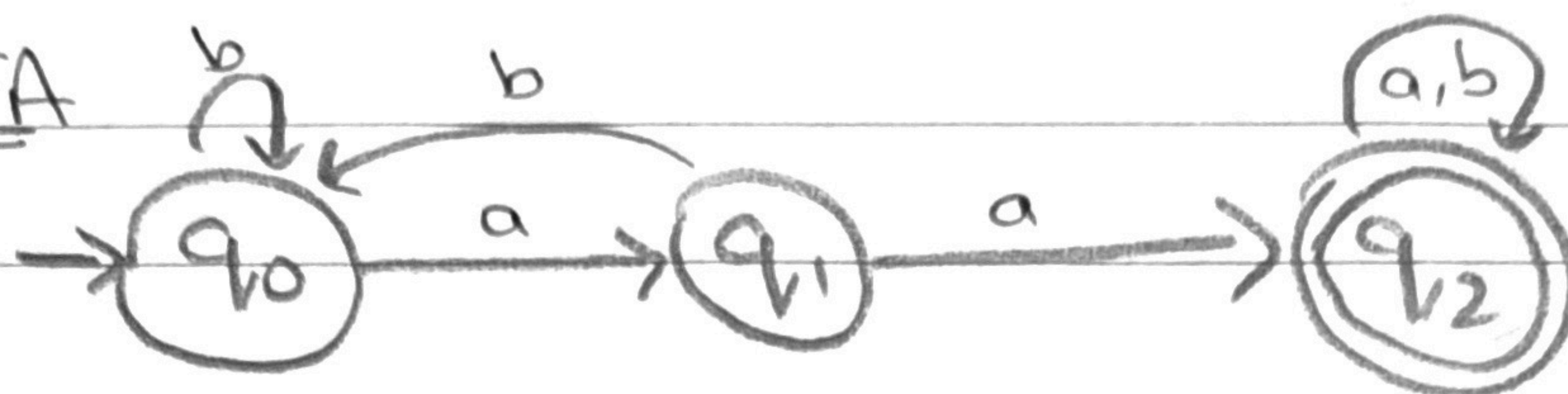
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Q71 $L_4 = \{ \text{All string Containing the Substring "aa"} \}$ Alphabet = $\{a, b\}$

NFA



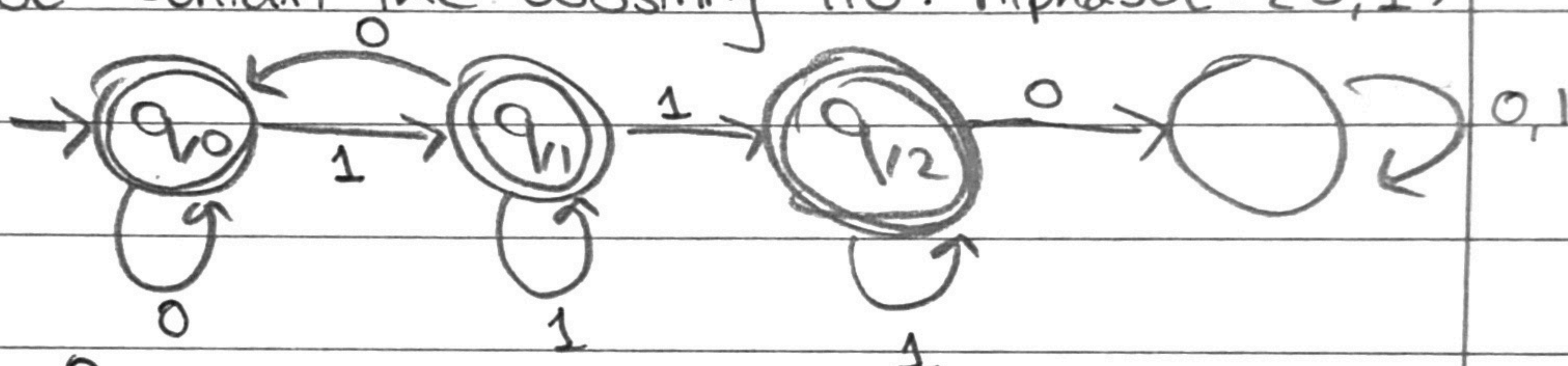
DFA



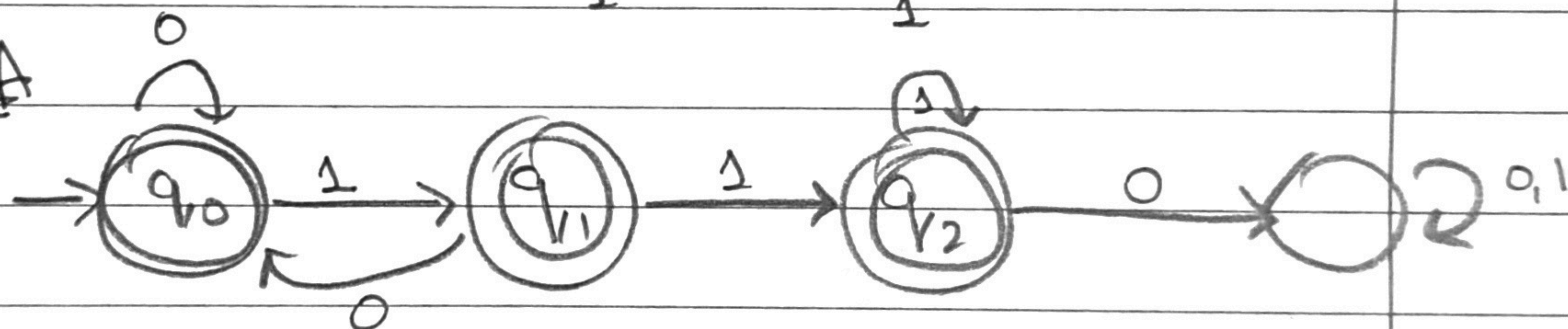
Section C

Q81 Draw a NFA, DFA, and TG that accept strings that do not contain the substring "110". Alphabet = $\{0, 1\}$

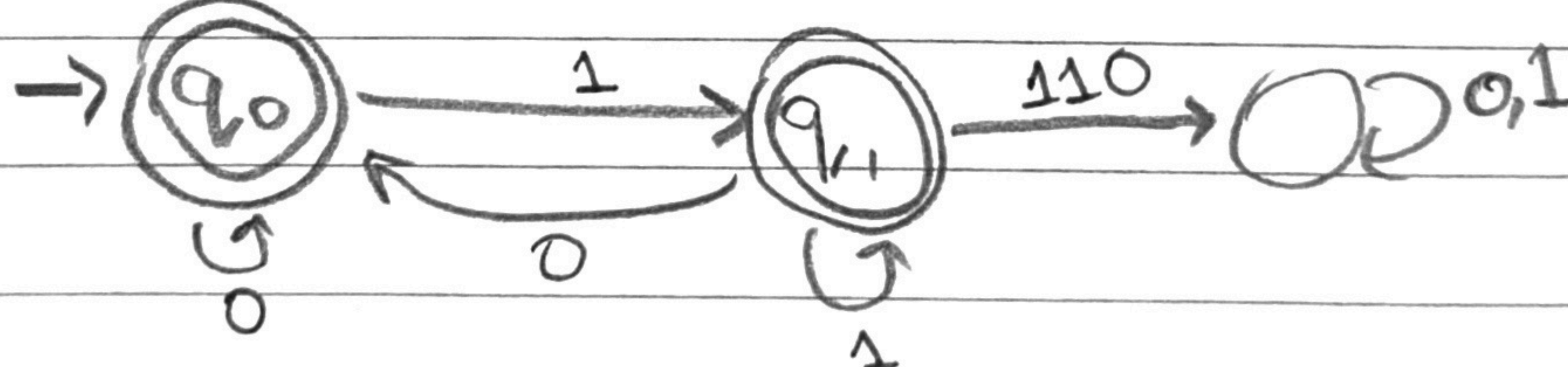
NFA



DFA



TG

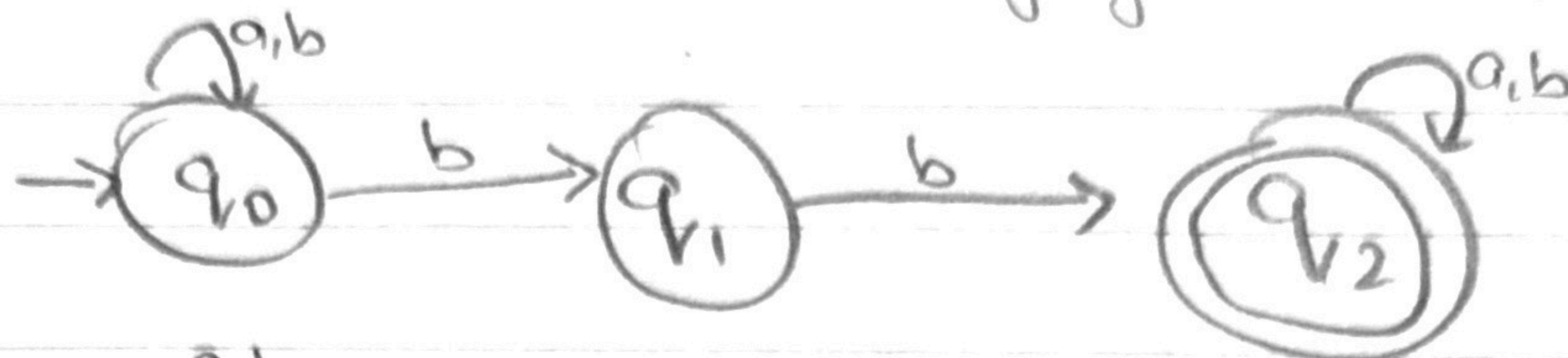


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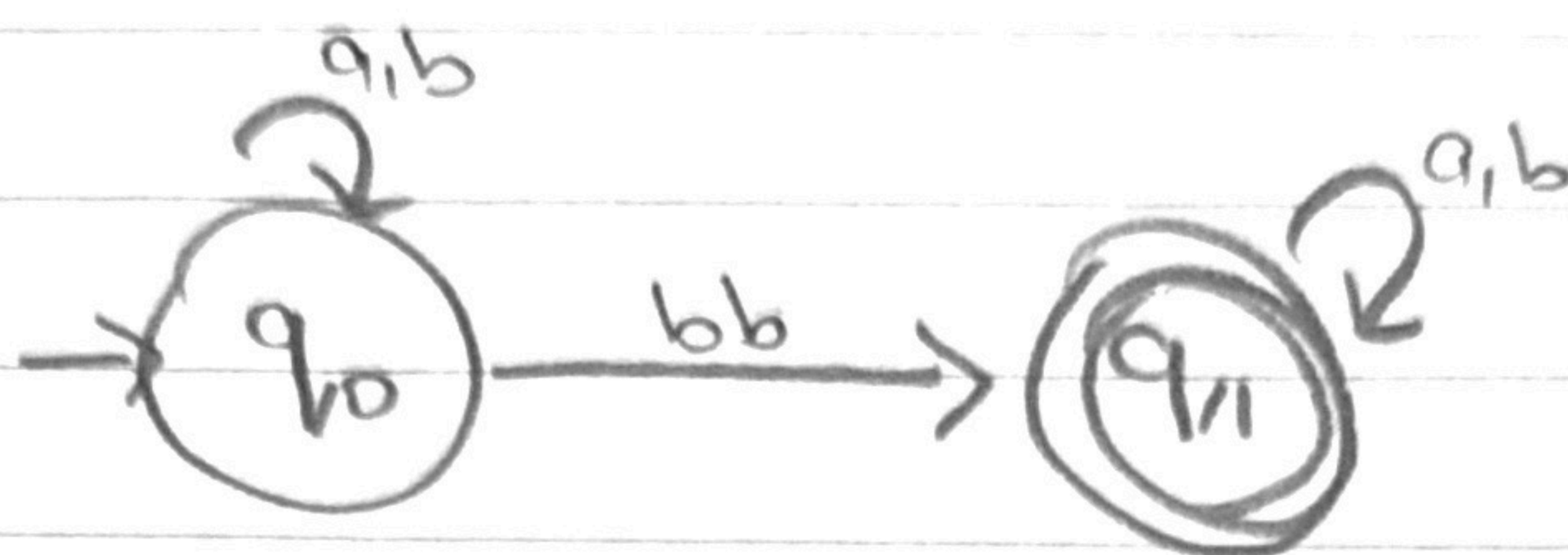
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Q9] Give an NFA and TG for the language: $(a+b)^*bb(a+b)^*$

NFA

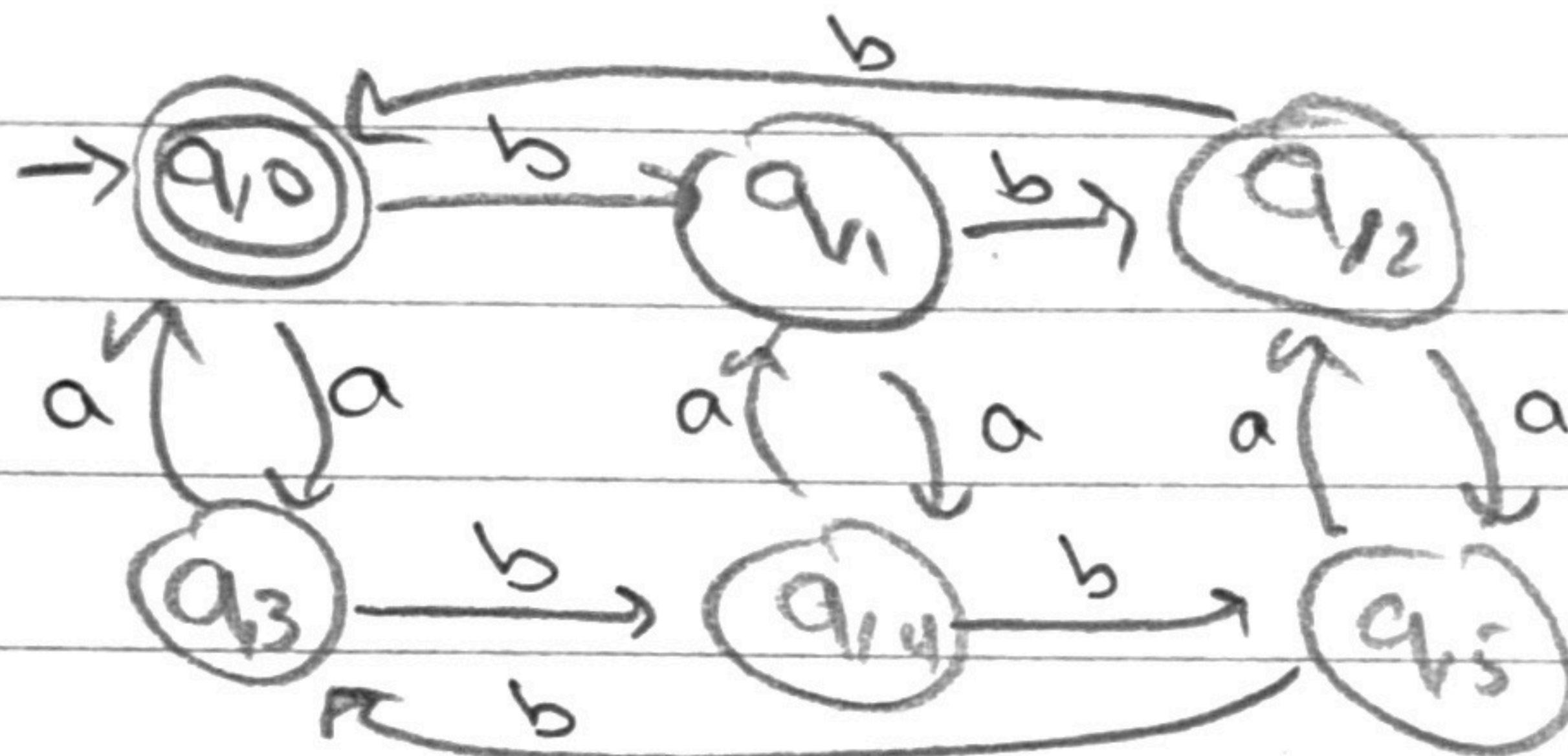


TG

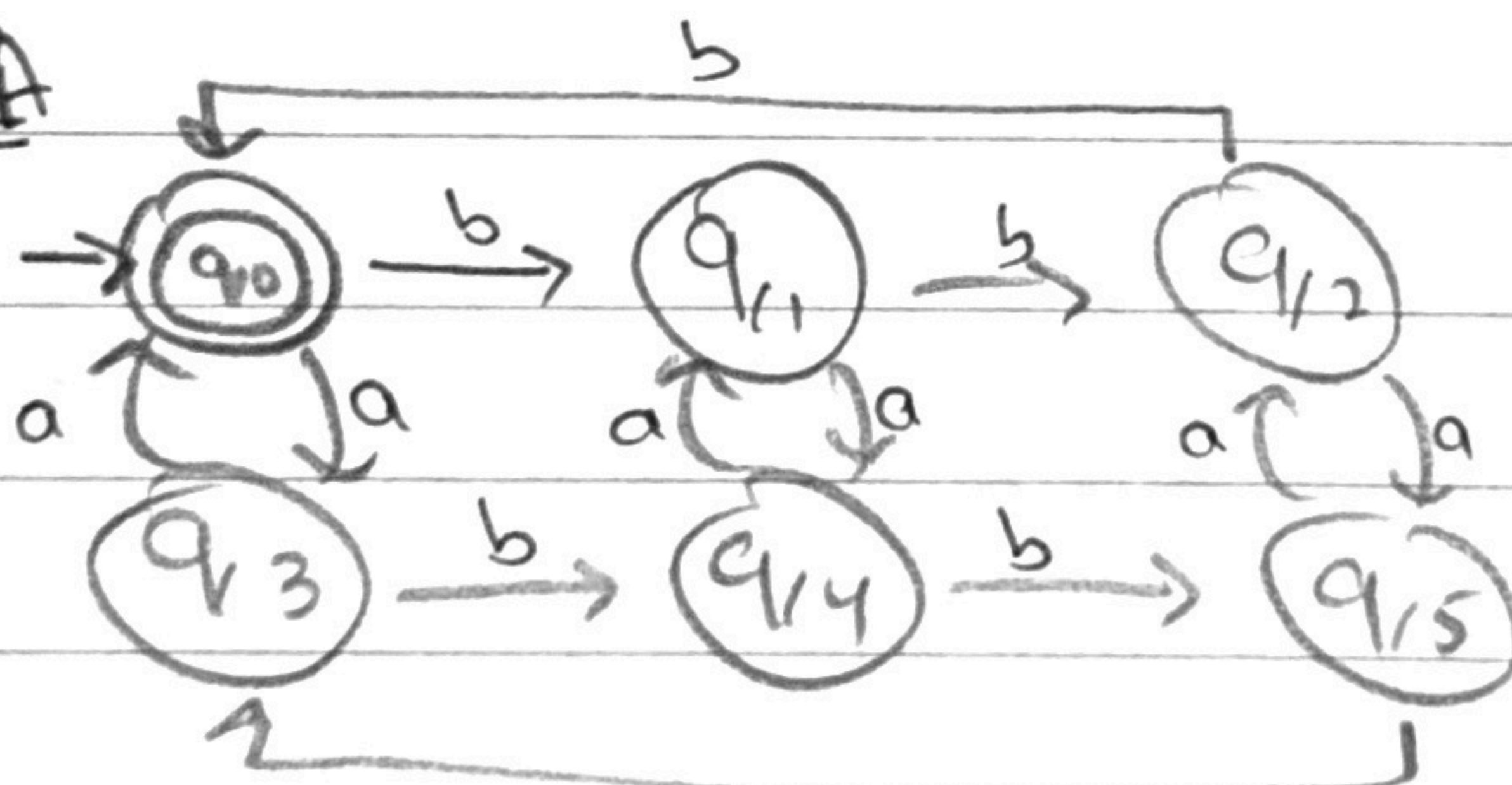


Q10] Design a NFA, DFA, and TG over $\Sigma = \{a, b\}$ that accepts strings where number of a's is even, AND Number of b's is a multiple of 3.

NFA



DFA



TG

